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## CHARACTER STRING DIVIDING APPARATUS, VOCABULARY GROUP GENERATING METHOD, AND STORAGE MEDIUM


### Abstract

A character string dividing apparatus includes processing circuitry. The processing circuitry generates a new vocabulary group based on an existing vocabulary group including a plurality of vocabularies each being associated with an identifier. The processing circuitry acquires a character string and divides the character string into a plurality of vocabularies based on at least one of the existing vocabulary group and the new vocabulary group. In generating the new vocabulary group, the processing circuitry determines whether a vocabulary of a plurality of vocabularies included in the new vocabulary group is commonly included in the existing and the new vocabulary group, and associates an embedding vector associated with the vocabulary included in the existing vocabulary group with the vocabulary included in the new vocabulary group without changing the embedding vector based on a determination indicating that the vocabulary is commonly included in the existing and the new vocabulary group.



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## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This patent application is based on and claims priority pursuant to 35 U.S.C. § 119(a) to Japanese Patent Application No. 2024-030164, filed on Feb. 29, 2024, in the Japan Patent Office, the entire disclosure of which is hereby incorporated by reference herein.

BACKGROUND

Technical Field

[0002] The present disclosure relates to a character string dividing apparatus, a vocabulary group generating method, and a storage medium.

Related Art

[0003] A technique has been proposed in which a vocabulary of a specific language is added to an existing model included in a tokenizer in order to cause the model to support the specific language.

SUMMARY

[0004] Embodiments of the present disclosure described herein provide a novel character string dividing apparatus includes processing circuitry. The processing circuitry generates a new vocabulary group based on an existing vocabulary group, the existing vocabulary group including a plurality of vocabularies each being associated with an identifier. The processing circuitry acquires a character string and divides the character string into a plurality of vocabularies based on at least one of the existing vocabulary group and the new vocabulary group. The processing circuitry outputs the plurality of vocabularies of the character string. In generating the new vocabulary group, the processing circuitry determines whether a vocabulary of a plurality of vocabularies included in the new vocabulary group is commonly included in the existing vocabulary group and the new vocabulary group, and associates an embedding vector associated with the vocabulary included in the existing vocabulary group with the vocabulary included in the new vocabulary group without changing the embedding vector based on a determination indicating that the vocabulary is commonly included in the existing vocabulary group and the new vocabulary group.

[0005] Embodiments of the present disclosure described herein provide a novel method for generating a vocabulary group. The method includes: generating a new vocabulary group based on an existing vocabulary group including a plurality of vocabularies each being associated with an identifier. The new vocabulary group is used when a character string is divided into a plurality of vocabularies. In generating the new vocabulary group, the method includes: determining whether a vocabulary among a plurality of vocabularies included in the new vocabulary group is commonly included in the existing vocabulary group and the new vocabulary group; and associating an embedding vector associated with the vocabulary included in the existing vocabulary group with the vocabulary included in the new vocabulary group without changing the embedding vector based on a determination indicating that the vocabulary is commonly included in the existing vocabulary group and the new vocabulary group.

[0006] Embodiments of the present disclosure described herein provide a novel non-transitory storage medium storing computer-readable program code that, when executed by a computer, causes the computer to perform a method for generating a vocabulary group. The method includes: generating a new vocabulary group based on an existing vocabulary group including a plurality of vocabularies each being associated with an identifier. The new vocabulary group is used when a character string is divided into a plurality of vocabularies. In generating the new vocabulary group, the method includes: determining whether a vocabulary among a plurality of vocabularies included in the new vocabulary group is commonly included in the existing vocabulary group and the new vocabulary group, and associating an embedding vector associated with the vocabulary included in the existing vocabulary group with the vocabulary included in the new vocabulary group without changing the embedding vector based on a determination indicating that the vocabulary is commonly included in the existing vocabulary group and the new vocabulary group.

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## Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] A more complete appreciation of embodiments of the present disclosure and many of the attendant advantages and features thereof can be readily obtained and understood from the following detailed description with reference to the accompanying drawings, wherein:

[0008] FIG. **1** is a diagram illustrating a system configuration of a character string dividing system according to an embodiment of the present disclosure;

[0009] FIG. **2** is a diagram illustrating a hardware configuration of a server according to an embodiment of the present disclosure;

[0010] FIG. **3** is a functional block diagram of a server according to an embodiment of the present disclosure;

[0011] FIG. **4** is another functional block diagram of a server according to an embodiment of the present disclosure;

[0012] FIG. **5** is a conceptual diagram illustrating character string dividing processing by a tokenizer included in a server according to an embodiment of the present disclosure;

[0013] FIG. **6** is a conceptual diagram illustrating generation processing performed by a generation unit included in a server according to an embodiment of the present disclosure;

[0014] FIG. **7** is a diagram illustrating generation processing performed by a generation unit included in a server according to an embodiment of the present disclosure;

[0015] FIG. **8** is a flowchart of a procedure of generation processing performed by a generation unit included in a server according to an embodiment of the present disclosure;

[0016] FIG. **9** is a diagram illustrating a first example of a vocabulary group included in a server according to an embodiment of the present disclosure;

[0017] FIG. **10** is a diagram illustrating a second example of a vocabulary group included in a server according to an embodiment of the present disclosure;

[0018] FIG. **11** is a flowchart of a procedure of vector setting processing performed by the generation unit included in the server according to an embodiment of the present disclosure;

[0019] FIG. **12** is a diagram illustrating vector setting processing performed by a generation unit included in a server according to an embodiment of the present disclosure;

[0020] FIG. **13** is a diagram illustrating a first example of a configuration of cooperation between a tokenizer and a machine learning model included in a server according to an embodiment of the present disclosure; and

[0021] FIG. **14** is a diagram illustrating a second example of a configuration of cooperation between a tokenizer and a machine learning model included in a server according to an embodiment of the present disclosure.

[0022] The accompanying drawings are intended to depict embodiments of the present disclosure and should not be interpreted to limit the scope thereof. The accompanying drawings are not to be considered as drawn to scale unless explicitly noted. Also, identical or similar reference numerals designate identical or similar components throughout the several views.

## DETAILED DESCRIPTION

[0023] In describing embodiments illustrated in the drawings, specific terminology is employed for the sake of clarity. However, the disclosure of this specification is not intended to be limited to the specific terminology so selected and it is to be understood that each specific element includes all technical equivalents that have a similar function, operate in a similar manner, and achieve a similar result.

[0024] Referring now to the drawings, embodiments of the present disclosure are described below. As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise.

[0025] A description is given below of embodiments of the present disclosure with reference to the drawings.

[0026] FIG. **1** is a diagram illustrating a system configuration of a character string dividing system **10** according to an embodiment of the present disclosure.

[0027] The character string dividing system **10** illustrated in FIG. **1** includes a server **12** and a personal computer (PC) **14**. The server **12** and the PC **14** are communicably connected to each other through a communication network **16**. The communication network **16** is, for example, a local area network (LAN) or the Internet.

[0028] The PC **14** is an example of a terminal apparatus. The PC **14** includes an input device for inputting a string of characters. Examples of the input device include, but not limited to, a keyboard and a touch screen. The PC **14** transmits a character string input by the user using the input device to the server **12** through the communication network **16**. The PC **14** may have a voice recognition function of recognizing a voice input from a microphone by the user. In this case, the PC **14** may transmit a character string recognized by the voice recognition function to the server **12**.

[0029] The server **12** is an example of a character string dividing apparatus. The server **12** receives the character string transmitted from the PC **14** through the communication network **16**, and performs the character string dividing processing to divide the character string into multiple vocabularies. The server **12** outputs the multiple vocabularies acquired by the character string dividing processing to the PC **14** through the communication network **16**.

[0030] The PC **14** receives the multiple vocabularies (i.e., the multiple vocabularies acquired by the character string dividing processing) transmitted from the server **12** through the communication network **16** and perform various processes using the multiple vocabularies.

[0031] In the example illustrated in FIG. **1**, one terminal apparatus (PC **14**) is connected to the server **12** through the communication network **16**. However, in actuality, in the character string dividing system **10**, multiple terminal apparatuses may be connected to the server **12** through the communication network **16**, and a character string may be transmitted to the server **12** from each of the multiple terminal apparatuses.

[0032] In the example illustrated in FIG. **1**, the PC **14** is used as an example of a terminal apparatus for inputting a character string. However, the terminal apparatus is not limited thereto, and may be another device (e.g., a smartphone, a tablet device, an electronic whiteboard, an inkjet printer, a multifunction peripheral (MFP), a video conference device, a projector, or a spherical imaging device) as long as the device can at least input and transmit a character string.

[0033] FIG. **2** is a diagram illustrating a hardware configuration of the server **12** and the PC **14** according to an embodiment of the present disclosure. A description is given below of the hardware configuration common to the server **12** and the PC **14**.

[0034] As illustrated in FIG. **2**, each of the server **12** and the PC **14** is implemented by a computer. The computer includes a central processing unit (CPU) **501**, a read-only memory (ROM) **502**, a random-access memory (RAM) **503**, a hard disk (HD) **504**, a hard disk drive (HDD) controller **505**, a display **506**, an external device connection interface (I/F) **508**, a network I/F **509**, a data bus **510**, a keyboard **511**, a pointing device **512**, a digital versatile disk rewritable (DVD-RW) drive **514**, and a medium I/F **516**.

[0035] The CPU **501** controls the overall operation of the server **12** or the PC **14**. The ROM **502** stores programs such as an initial program loader (IPL) to boot the CPU **501**. The RAM **503** is used as a work area for the CPU **501**. The HD **504** stores various data such as a program. The HDD controller **505** controls the reading and writing of various data from and to the HD **504** under the control of the CPU **501**. The display **506** displays various information such as a cursor, a menu, a window, a character, or an image. The external device connection I/F **508** is an interface for connecting the computer **500** to various external devices. In this case, the external devices include, but not limited to, a universal serial bus (USB) memory and a printer. The network I/F **509** is an interface that controls data communication performed with an external device through the communication network **16**. The data bus **510** is, for example, an address bus or a data bus, which electrically couples the components illustrated in FIG. **2**, such as the CPU **501**.

[0036] The keyboard **511** is an input device provided with multiple keys for allowing the user to input characters, numerals, or various instructions. The pointing device **512** is an example of an input device that allows the user to select or execute a specific instruction, select a target for processing, or move a cursor being displayed. The DVD-RW drive **514** controls reading and writing of various data from and to a DVD-RW **513**, which is an example of a removable storage medium. The DVD-RW is one example of the removable storage medium. In another example, a digital versatile disk recordable (DVD-R) may be used as the removable storage medium. The medium I/F **516** controls reading and writing (storing) of data from and to a storage medium **515** such as a flash memory.

[0037] FIGS. **3** and **4** are functional block diagrams of the server **12** according to an embodiment of the present disclosure. FIG. **3** is the functional block diagram of the server **12** before a tokenizer **110-2** is generated. FIG. **4** is the functional block diagram of the server **12** after the tokenizer **110-2** has been generated.

[0038] As illustrated in FIG. **3**, the server **12**, which is in a state before the tokenizer **110-2** is generated, includes an acquisition unit **101**, an output unit **102**, a generation unit **103**, and a tokenizer **110-1**.

[0039] The acquisition unit **101** acquires a character string that is transmitted from the PC **14** and received by the server **12** through the communication network **16** as a character string to be processed by the character string dividing processing.

[0040] The tokenizer **110-1** is an example of a character string dividing unit. The tokenizer **110-1** includes a vocabulary group **111-1**. The vocabulary group **111-1** is an example of an existing vocabulary group. The vocabulary group **111-1** includes multiple vocabularies associated with identifiers. The vocabulary group **111-1** is implemented by, for example, a JavaScript® object notation (JSON) file. The tokenizer **110-1** performs the character string dividing processing (i.e., tokenization processing) using an existing technique on the character string acquired by the acquisition unit **101** based on the vocabulary group **111-1** to divide the character string acquired by the acquisition unit **101** into multiple vocabularies (i.e., multiple morphemes).

[0041] The output unit **102** outputs the multiple vocabularies acquired by the tokenizer **110-1**. The multiple vocabularies outputted by the output unit **102** are transmitted to the PC **14** by the server **12** through the communication network **16**.

[0042] The generation unit **103** generates a new tokenizer **110**. For example, the generation unit **103** generates a tokenizer **110-2** as illustrated in FIG. **4** as a new tokenizer **110** for the server **12** including the tokenizer **110-1** as illustrated in FIG. **3**. At this time, the generation unit **103** generates a vocabulary group **111-2** included in the tokenizer **110-2** based on the vocabulary group **111-1** included in the tokenizer **110-1**.

[0043] As illustrated in FIG. **4**, the server **12**, which is in a state after the tokenizer **110-2** has been generated, further includes the tokenizer **110-2** in addition to the server **12** illustrated in FIG. **3**.

[0044] The tokenizer **110-2** is another example of the character string dividing unit. The tokenizer **110-2** includes the vocabulary group **111-2**. The vocabulary group **111-2** is an example of a new vocabulary group. The vocabulary group **111-2** includes multiple vocabularies associated with identifiers. The vocabulary group **111-2** is implemented by, for example, a JSON file. The tokenizer **110-2** performs the character string dividing processing using an existing technique on the character string acquired by the acquisition unit **101** based on the vocabulary group **111-2** to divide the character string acquired by the acquisition unit **101** into multiple vocabularies (i.e., multiple morphemes).
[0045] The server **12** illustrated in FIG. **4** includes the tokenizer **110-1** and the tokenizer **110-2**. Accordingly, the server **12** illustrated in FIG. **4** allows the user to select either the tokenizer **110-1** or the tokenizer **110-2** as the tokenizer to be used for the character string dividing processing.
[0046] Alternatively, the server **12** illustrated in FIG. **4** may automatically select one of the tokenizer **110-1** and the tokenizer **110-2** as the tokenizer to be used for the character string dividing processing based on a predetermined determination condition (e.g., whether the character string acquired by the acquisition unit **101** is English or Japanese).
[0047] Alternatively, the server **12** illustrated in FIG. **4** may use the new tokenizer **110-2** as the tokenizer used for the character string dividing processing and may not use the existing tokenizer **110-1**.
[0048] In the server **12** illustrated in FIG. **4**, when the tokenizer **110-1** is selected, the tokenizer **110-1** performs the character string dividing processing on the character string acquired by the acquisition unit **101** based on the vocabulary group **111-1**.
[0049] In the server **12** illustrated in FIG. **4**, when the tokenizer **110-2** is selected, the tokenizer **110-2** performs the character string dividing processing on the character string acquired by the acquisition unit **101** based on the vocabulary group **111-2**.
[0050] FIG. **5** is a conceptual diagram illustrating the character string dividing processing by the tokenizer **110** included in the server **12** according to an embodiment of the present disclosure.
[0051] As illustrated in FIG. **5**, the tokenizer **110** includes a vocabulary group **111**, an encoder **112**, and a decoder **113**.
[0052] The vocabulary group **111** includes multiple vocabularies associated with identifiers.
[0053] The encoder **112** performs the character string dividing processing on the input character string based on the vocabulary group **111** to divide the input character string into multiple vocabularies and output the multiple vocabularies.
[0054] For example, in the example illustrated in FIG. **5**, when a character string “is going to be transmitted” is input as an input example **1**, the encoder **112** divides the character string “is going to be transmitted” into the vocabulary “is going to be” and the vocabulary “transmitted” based on the vocabulary group **111**, and outputs an identifier string “[7580, 50785]” including identifiers of the multiple vocabularies as an output example 1.
[0055] The decoder **113** performs character string combining processing on the multiple input vocabularies based on the vocabulary group **111** to combine the multiple input vocabularies to generate a character string and output the character string.
[0056] For example, in the example illustrated in FIG. **5**, when the identifier string “[7580, 40833]”is input as an input example 2, the decoder **113** combines the vocabulary “returned” corresponding to the identifier [40833] and the vocabulary “is going to be” corresponding to the identifier [7580] based on the vocabulary group **111** to generate a character string “is going to be returned,” and outputs the character string “is going to be returned” as an output example 2.
[0057] FIG. **6** is a conceptual diagram illustrating generation processing performed by the generation unit **103** included in the server **12** according to an embodiment of the present disclosure.
[0058] As illustrated in FIG. **6**, when the generation unit **103** generates the tokenizer **110-2**, the generation unit **103** generates the new vocabulary group **111-2** included in the tokenizer **110-2** based on the existing vocabulary group **111-1** included in the tokenizer **110-1**.

[0059] Specifically, the generation unit **103** copies the vocabularies included in a common vocabulary group **111-2**A that includes the vocabularies commonly included in the existing vocabulary group **111-1** and the new vocabulary group **111-2** to the new vocabulary group **111-2** without changing the identifiers.

[0060] The generation unit **103** assigns any identifiers to vocabularies that are not included in the common vocabulary group **111-2**A with the existing vocabulary group **111-1** (i.e., vocabularies included in a non-common vocabulary group **111-2**B with the existing vocabulary group **111-1**) among the multiple vocabularies included in the new vocabulary group **111-2** and adds the vocabularies to the new vocabulary group **111-2**.

[0061] Thus, the generation unit **103** can efficiently generate the new vocabulary group **111-2** based on the existing vocabulary group **111-1** without newly generating the new vocabulary group **111-2**.

[0062] When the generation unit **103** generates the new vocabulary group **111-2**, the generation unit **103** does not include a non-common vocabulary group **111-1**A of the existing vocabulary group **111-1** in the new vocabulary group **111-2**. In other words, the generation unit **103** does not simply add the non-common vocabulary group **111-2**B to the existing vocabulary group **111-1**. Accordingly, the generation unit **103** can reduce the expansion of the data size of the new vocabulary group **111-2**.

[0063] Since the generation unit **103** reuses the identifiers used for the non-common vocabulary group **111-1**A of the existing vocabulary group **111-1** for the non-common vocabulary group **111-2**B of the new vocabulary group **111-2**, the generation unit **103** does not need to assign new identifiers. As a result, the generation unit **103** can reduce the expansion of the data size of the new vocabulary group **111-2**.

[0064] FIG. **7** is a diagram illustrating the generation processing performed by the generation unit **103** included in the server **12** according to an embodiment of the present disclosure.

[0065] As illustrated in FIG. **7**, the generation unit **103** copies the vocabulary “this” and the vocabulary “important” included in the common vocabulary group **111-2**A, which is common to the new vocabulary group **111-2** and the existing vocabulary group **111-1** among the multiple vocabularies included in the new vocabulary group **111-2**, to the new vocabulary group **111-2** without changing the identifiers (“200” and “202”). Thus, the generation unit **103** can efficiently generate the new vocabulary group **111-2** based on the existing vocabulary group **111-1** without newly generating the new vocabulary group **111-2**.

[0066] As illustrated in FIG. **7**, the generation unit **103** adds the vocabulary “tomorrow” and the vocabulary “hello”, which are not included in the common vocabulary group **111-2**A with the existing vocabulary group **111-1**, among the multiple vocabularies included in the new vocabulary group **111-2** to the new vocabulary group **111-2** and reuses the identifiers (“201” and “203”) used in the non-common vocabulary group **111-1**A of the existing vocabulary group **111-1** (i.e., replacing the vocabularies of the identifiers). Accordingly, the generation unit **103** does not need to assign new identifiers, and thus, the generation unit **103** can reduce the expansion of the data size of the new vocabulary group **111-2**.

[0067] In other words, the generation unit **103** can prevent the number of identifiers of the new vocabulary group **111-2** from increasing over the number of identifiers of the existing vocabulary group **111-1**. As a result, the generation unit **103** can prevent the number of identifiers of a large language model (LLM) corresponding to the new vocabulary group **111-2** from increasing over the number of identifiers of the LLM corresponding to the existing vocabulary group **111-1**, and thus, the generation unit **103** can reduce the increasing of memory cost related to the LLM corresponding to the new vocabulary group **111-2**.

[0068] FIG. **8** is a flowchart of a procedure of the generation processing performed by the generation unit **103** included in the server **12** according to an embodiment of the present disclosure.

[0069] In step S**801**, the generation unit **103** selects one vocabulary from among multiple vocabularies included in the new vocabulary group **111-2**.

[0070] Subsequently, in step S**802**, the generation unit **103** determines whether the vocabulary selected in step S**801** is included in the existing vocabulary group **111-1**.
[0071] In step S**803**, when the generation unit **103** determines that the vocabulary selected in step S**801** is included in the existing vocabulary group **111-1** in step S**802** (YES in step S**802**), the generation unit **103** copies the vocabulary selected in step S**801** to the new vocabulary group **111-2** without changing the identifier of the vocabulary. Then, the generation unit **103** proceeds the operation to step S**806**.
[0072] In step S**804**, when it is determined that the vocabulary selected in step S**801** is not included in the existing vocabulary group **111-1** in step S**802** (NO in step S**802**), the generation unit **103** assigns the identifier used in the non-common vocabulary group **111-1**A of the existing vocabulary group **111-1** to the vocabulary selected in step S**801**. Then, in step S**805**, the generation unit **103** registers the vocabulary selected in step S**801** and the identifier assigned in step S**804** in the new vocabulary group **111-2**. Then, the generation unit **103** proceeds the operation to step S**806**.
[0073] In step S**806**, the generation unit **103** determines whether all of the multiple vocabularies included in the new vocabulary group **111-2** have been selected.
[0074] When the generation unit **103** determines that all of the multiple vocabularies included in the new vocabulary group **111-2** have not been selected in step S**806** (NO in step S**806**), the generation unit **103** returns the operation to step S**801**.
[0075] When the generation unit **103** determines that all of the multiple vocabularies included in the new vocabulary group **111-2** have been selected in step S**806** (YES in step S**806**), the generation unit **103** ends the series of processing illustrated in FIG. **8**.
[0076] FIG. **9** is a diagram illustrating a first example of the vocabulary group **111-1** and the vocabulary group **111-2** included in the server **12** according to an embodiment of the present disclosure.
[0077] In the example illustrated in FIG. **9**, an English vocabulary group including multiple English vocabularies (dog, cat, this, and day) is used as the existing vocabulary group **111-1**.
[0078] In the example illustrated in FIG. **9**, an English vocabulary group including multiple Japanese vocabularies (inu, neko, kore, and hi) is used as the new vocabulary group **111-2**.
[0079] In other words, in the example illustrated in FIG. **9**, the existing vocabulary group **111-1** and the new vocabulary group **111-2** are generated in units of languages.
[0080] FIG. **10** is a diagram illustrating a second example of the vocabulary group **111-1** and the vocabulary group **111-2** included in the server **12** according to an embodiment of the present disclosure.
[0081] In the example illustrated in FIG. **10**, a vocabulary group of daily terms including vocabularies of multiple daily terms (walk, bucket, and blanket) is used as the existing vocabulary group **111-1**.
[0082] In the example illustrated in FIG. **10**, a vocabulary group of business terms including vocabularies of multiple business terms (gross profit, business, and KPI) is used as the new vocabulary group **111-2**.
[0083] In other words, in the example illustrated in FIG. **10**, the existing vocabulary group **111-1** and the new vocabulary group **111-2** are generated in units of terms.
[0084] FIG. **11** is a flowchart of a procedure of vector setting processing performed by the generation unit **103** included in the server **12** according to an embodiment of the present disclosure. For example, the generation unit **103** performs the series of processing illustrated in FIG. **11** after the generation unit **103** generates the new vocabulary group **111-2**.
[0085] In step S**1101**, the generation unit **103** selects one vocabulary from among multiple vocabularies included in the new vocabulary group **111-2**.
[0086] Subsequently, in step S**1102**, the generation unit **103** determines whether the vocabulary selected in step S**1101** is included in the existing vocabulary group **111-1**.
[0087] In step S**1103**, when the generation unit **103** determines that the vocabulary selected in step

S**1101** is included in the existing vocabulary group **111-1** in step S**1102** (YES in step S**1102**), the generation unit **103** sets an embedding vector associated with the vocabulary in an existing LLM **120-1** to the identifier of the vocabulary selected in step S**1101** in a new LLM **120-2** without changing the embedding vector. Then, the generation unit **103** proceeds the operation to step S**1106**.
[0088] In step S**1104**, when the generation unit **103** determines that the vocabulary selected in step S**1101** is not included in the existing vocabulary group **111-1** in step S**1102** (NO in step S**1102**), the generation unit **103** divides the vocabulary selected in step S**1101** into multiple vocabularies based on the existing vocabulary group **111-1** using the existing tokenizer **110-1**. In step S**1105**, the generation unit **103** sets the average value of the multiple embedding vectors associated with the multiple vocabularies acquired in step S**1104** in the existing LLM **120-1** to the identifier of the vocabulary selected in step S**1101** in the new LLM **120-2**. Then, the generation unit **103** proceeds the operation to step S**1106**.
[0089] In step S**1106**, the generation unit **103** determines whether all of the multiple vocabularies included in the new vocabulary group **111-2** have been selected.
[0090] When the generation unit **103** determines that all of the multiple vocabularies included in the new vocabulary group **111-2** have not been selected in step S**1106** (NO in step S**1106**), the generation unit **103** returns the operation to step S**1101**.
[0091] When the generation unit **103** determines that all of the multiple vocabularies included in the new vocabulary group **111-2** have been selected in step S**1106** (YES in step S**1106**), the generation unit **103** ends the series of processing illustrated in FIG. **8**.
[0092] FIG. **12** is a diagram illustrating vector setting processing performed by the generation unit **103** included in the server **12** according to an embodiment of the present disclosure.
[0093] As illustrated in FIG. **12**, embedding vectors are associated with multiple vocabularies of the existing vocabulary group **111-1** in one-to-one correspondence. Specifically, the embedding vectors of the multiple vocabularies of the existing vocabulary group **111-1** are provided in the existing LLM **120-1** corresponding to the existing vocabulary group **111-1**, and are associated with the vocabularies of the existing vocabulary group **111-1** by the identifiers.
[0094] Similarly, the embedding vectors are associated with the multiple vocabularies of the new vocabulary group **111-2**, in one-to-one correspondence. Specifically, the embedding vectors of the multiple vocabularies of the new vocabulary group **111-2** are provided in the new LLM **120-2** corresponding to the new vocabulary group **111-2**, and are associated with the vocabularies of the existing vocabulary group **111-1** by the identifiers.
[0095] In the example illustrated in FIG. **12**, the generation unit **103** copies a vocabulary “mail”, which is included in the existing vocabulary group **111-1** among the multiple vocabularies included in the new vocabulary group **111-2**, to the new vocabulary group **111-2** without changing an identifier “2549”. At this time, the generation unit **103** associates the embedding vector associated with the identifier “2549” of the vocabulary “mail” in the existing LLM **120-1** with the identifier “2549” of the vocabulary “mail” in the new LLM **120-2** without changing the embedding vector. As a result, the generation unit **103** can efficiently set the embedding vector of the vocabulary common to the existing vocabulary group **111-1** in the new LLM **120-2**.
[0096] In the example illustrated in FIG. **12**, the generation unit **103** adds a vocabulary “queen”, which is not included in the existing vocabulary group **111-1** among the multiple vocabularies included in the new vocabulary group **111-2**, to the new vocabulary group **111-2** and reuses the identifier “13603” used for the vocabulary “XXX” of the non-common vocabulary group **111-1**A of the existing vocabulary group **111-1** (i.e., replacing the vocabulary of the identifier). At this time, the generation unit **103** uses the existing tokenizer **110-1** to divide the vocabulary “queen” into a vocabulary “woman” and a vocabulary “king” based on the existing vocabulary group **111-1**. Then, the generation unit **103** associates the average value of the embedding vector associated with the identifier “30647” of the existing vocabulary “woman” in the LLM **120-1** and the embedding

vector associated with the identifier "30462" of the existing vocabulary "king" in the LLM **120-1** with the identifier "13603" of the vocabulary "queen" in the new LLM **120-2**. As a result, the generation unit **103** can efficiently generate and set the embedding vector of the vocabulary that is not common to the existing vocabulary group **111-1** for the new LLM **120-2**.

[0097] FIG. **13** is a diagram illustrating a first example of a configuration of cooperation between the tokenizer **110** and a machine learning model included in the server **12** according to an embodiment of the present disclosure. FIG. **13** illustrates a configuration in which the tokenizer **110** cooperates with an LLM **120**, which is an example of the machine learning model. The LLM **120** may be provided in the server **12** or in a device outside the server **12**.

[0098] As illustrated in FIG. **13**, the LLM **120** includes an embedded layer **121**, a hidden layer **122**, and an output layer **123**.

[0099] In the example illustrated in FIG. **13**, in step (1), a character string is input to the tokenizer **110**. In step (2), the tokenizer **110** converts the input character string into an identifier string based on the vocabulary group **111**, and outputs the identifier string to the LLM **120**. In step (3), the LLM **120** generates an identifier string following the input identifier string and outputs the identifier string to the tokenizer **110**. In step (4), the tokenizer **110** converts the identifier string input from the LLM **120** into a character string based on the vocabulary group **111**, and outputs the character string.

[0100] FIG. **14** is a diagram illustrating a second example of a configuration of cooperation between the tokenizer **110** and the machine learning model included in the server **12** according to an embodiment of the present disclosure. FIG. **14** illustrates the configuration in which the tokenizer **110** cooperates with a text-to-image model **130**, which is another example of the machine learning model. The text-to-image model **130** may be provided in the server **12** or in an external device of the server **12**.

[0101] As illustrated in FIG. **14**, the text-to-image model **130** includes an embedded layer **131**, a noise generation unit **132**, a noise removal model **133**, and an output layer **134**.

[0102] In step (1) in FIG. **14**, a character string is input to the tokenizer **110**. In step (2), the tokenizer **110** converts the input character string into an identifier string based on the vocabulary group **111**, and outputs the identifier string to the text-to-image model **130**. In step (3), the text-to-image model **130** generates an image based on the input identifier string, and outputs the image.

[0103] In FIGS. **13** and **14**, the LLM **120** and the text-to-image model **130** are used as examples of the machine learning model. However, the present disclosure is not limited thereto, and image-to-text or music generation artificial intelligence may be used as the machine learning model.

[0104] Although some embodiments of the present disclosure have been described in detail above, the present disclosure is not limited to such specific embodiments, and various modifications and changes can be made within the scope of the gist of the invention described in the claims.

[0105] The group of apparatuses or devices according to the embodiments of the present disclosure are merely one example of multiple computing environments that implement the embodiments disclosed in the present specification. In some embodiments, the management server **12** includes multiple computing devices, such as a server cluster. The multiple computing devices are configured to communicate with one another through any type of communication link including, for example, a network or a shared memory, and perform the processes disclosed in the present specification.

[0106] The above-described embodiments are illustrative and do not limit the present invention. Thus, numerous additional modifications and variations are possible in light of the above teachings. For example, elements and/or features of different illustrative embodiments may be combined with each other and/or substituted for each other within the scope of the present invention. Any one of the above-described operations may be performed in various other ways, for example, in an order different from the one described above.

[0107] The functionality of the elements disclosed herein may be implemented using circuitry or processing circuitry which includes general purpose processors, special purpose processors, integrated circuits, application-specific integrated circuits (ASICs), field-programmable gate arrays (FPGAs), and/or combinations thereof which are configured or programmed, using one or more programs stored in one or more memories, to perform the disclosed functionality. Processors are considered processing circuitry or circuitry as they include transistors and other circuitry therein. In the disclosure, the circuitry, units, or means are hardware that carry out or are programmed to perform the recited functionality. The hardware may be any hardware disclosed herein which is programmed or configured to carry out the recited functionality.

[0108] There is a memory that stores a computer program which includes computer instructions. These computer instructions provide the logic and routines that enable the hardware (e.g., processing circuitry or circuitry) to perform the method disclosed herein. This computer program can be implemented in known formats as a computer-readable storage medium, a computer program product, a memory device, a record medium such as a CD-ROM or DVD, and/or the memory of an FPGA or ASIC.

## Claims

**1**. A character string dividing apparatus comprising: processing circuitry configured to: generate a new vocabulary group based on an existing vocabulary group, the existing vocabulary group including a plurality of vocabularies each being associated with an identifier; acquire a character string; divide the character string into a plurality of vocabularies based on at least one of the existing vocabulary group and the new vocabulary group; and output the plurality of vocabularies of the character string, wherein, in generating the new vocabulary group, the processing circuitry is configured to: determine whether a vocabulary of a plurality of vocabularies included in the new vocabulary group is commonly included in the existing vocabulary group and the new vocabulary group; and associate an embedding vector associated with the vocabulary included in the existing vocabulary group with the vocabulary included in the new vocabulary group without changing the embedding vector based on a determination indicating that the vocabulary is commonly included in the existing vocabulary group and the new vocabulary group.

**2**. The character string dividing apparatus according to claim 1, wherein, among the plurality of vocabularies included in the new vocabulary group, the processing circuitry is configured to copy a vocabulary commonly included in the existing vocabulary group and the new vocabulary group to the new vocabulary group without changing the identifier, and the processing circuitry is configured to assign a new identifier to a vocabulary that is included in the new vocabulary group but not included in the existing vocabulary group and adds the vocabulary to the new vocabulary group.

**3**. The character string dividing apparatus according to claim 1, wherein, in a case where a first vocabulary included in the new vocabulary group is not included in the existing vocabulary group, the processing circuitry is configured to assign an identifier of a second vocabulary included in the existing vocabulary group but not included in the new vocabulary group to the first vocabulary included in the new vocabulary group, and add the first vocabulary to the new vocabulary group.

**4**. The character string dividing apparatus according to claim 1, wherein, among the plurality of vocabularies included in the new vocabulary group, the processing circuitry is configured to divide a vocabulary not commonly included in the existing vocabulary group into a plurality of vocabularies based on the existing vocabulary group, and associate an average value of a plurality of embedding vectors associated with the plurality of vocabularies included in the existing vocabulary group with the vocabulary of the new vocabulary group.

**5**. The character string dividing apparatus according to claim 1, wherein the existing vocabulary group and the new vocabulary group are generated in units of languages.

**6**. The character string dividing apparatus according to claim 1, wherein the existing vocabulary group and the new vocabulary group are generated in units of terms.

**7**. The character string dividing apparatus according to claim 1, wherein the processing circuitry is configured to output the identifiers of the plurality of vocabularies included in at least one of the existing vocabulary group and the new vocabulary group to a machine learning model and acquire a processing result from the machine learning model for the identifiers of the plurality of vocabularies included in the at least one of the existing vocabulary group and the new vocabulary group.

**8**. A method for generating a vocabulary group, the method comprising: generating a new vocabulary group based on an existing vocabulary group including a plurality of vocabularies each being associated with an identifier, the new vocabulary group being used when a character string is divided into a plurality of vocabularies, wherein the generating the new vocabulary group includes: determining whether a vocabulary among a plurality of vocabularies included in the new vocabulary group is commonly included in the existing vocabulary group and the new vocabulary group; and associating an embedding vector associated with the vocabulary included in the existing vocabulary group with the vocabulary included in the new vocabulary group without changing the embedding vector based on a determination indicating that the vocabulary is commonly included in the existing vocabulary group and the new vocabulary group.

**9**. A non-transitory storage medium storing computer-readable program code that, when executed by a computer, causes the computer to perform a method for generating a vocabulary group, the method comprising: generating a new vocabulary group based on an existing vocabulary group including a plurality of vocabularies each being associated with an identifier, the new vocabulary group being used when a character string is divided into a plurality of vocabularies, wherein, the generating the new vocabulary group includes: determining whether a vocabulary among a plurality of vocabularies included in the new vocabulary group is commonly included in the existing vocabulary group and the new vocabulary group, and associating an embedding vector associated with the vocabulary included in the existing vocabulary group with the vocabulary included in the new vocabulary group without changing the embedding vector based on a determination indicating that the vocabulary is commonly included in the existing vocabulary group and the new vocabulary group.